

Final report of data engineering project

(LinkedIn Posts)

Executive Summary

This project delves into the construction of an Extract, Transform, Load (ETL) process and the establishment of a data warehouse to facilitate in-depth analysis of LinkedIn posts and the companies behind them. By extracting data from LinkedIn's public API that was collected in Kaggle website, the project aims to create a centralized repository of valuable information for various stakeholders, including job seekers, recruiters, businesses, journalists, and researchers.

Introduction

Project Overview

Every day, thousands of companies and individuals turn to LinkedIn in search of talent. This dataset contains a nearly comprehensive record of 124,000+ job postings listed in 2023 and 2024. Each individual posting contains dozens of valuable attributes for both postings and companies, including the title, job description, salary, location, application URL, and work-types (remote, contract, etc), in addition to separate files containing the benefits, skills, and industries associated with each posting. The majority of jobs are also linked to a company, which are all listed in another csv file containing attributes such as the company description, headquarters location, and number of employees, and follower count.

The project focuses on harvesting and analyzing LinkedIn posts to gain insights into job market trends, company hiring practices, and in-demand skills. The extracted data will be transformed and loaded into a data warehouse, enabling comprehensive analysis and visualization.

Analytical Goals

The project seeks to achieve the following analytical objectives:

- **Identify the most sought-after skills and experience for various job roles:** Analyze LinkedIn posts to identify the most frequently mentioned skills and experience requirements for different job titles.
- **Understand the prevalent compensation structures for different positions:** Analyze LinkedIn posts to gain insights into the average salary ranges, benefits packages, and other compensation details for various job roles.
- **Explore company hiring trends and preferences:** Analyze LinkedIn posts to identify the companies actively hiring, their preferred job titles, and the skills they seek.
- **Uncover industry-specific job market trends:** Analyze LinkedIn posts to identify the most in-demand skills, common job titles, and prevalent compensation structures across different industries.

Data Sources

The project will primarily utilize data from LinkedIn's public API, which provides access to a vast repository of job posts, company profiles, and user information.

The file contains 11 CSV files, each pertaining to a specific section. Among these 11 files, there is a main file named posting.csv to which 6 other files are related. Additionally, there is

another file named companies.csv to which 3 files are related. Below, the details of each field within these files are explained in full:

job_postings.csv

job_id: The job ID as defined by LinkedIn ([https://www.linkedin.com/jobs/view/ job_id](https://www.linkedin.com/jobs/view/job_id))

company_id: Identifier for the company associated with the job posting (maps to companies.csv)

title: Job title.

description: Job description.

max_salary: Maximum salary

med_salary: Median salary

min_salary: Minimum salary

pay_period: Pay period for salary (Hourly, Monthly, Yearly)

formatted_work_type: Type of work (Fulltime, Parttime, Contract)

location: Job location

applies: Number of applications that have been submitted

original_listed_time: Original time the job was listed

remote_allowed: Whether job permits remote work

views: Number of times the job posting has been viewed

job_posting_url: URL to the job posting on a platform

application_url: URL where applications can be submitted

application_type: Type of application process (offsite, complex/simple onsite)

expiry: Expiration date or time for the job listing

closed_time: Time to close job listing

formatted_experience_level: Job experience level (entry, associate, executive, etc)

skills_desc: Description detailing required skills for job

listed_time: Time when the job was listed

posting_domain: Domain of the website with application

sponsored: Whether the job listing is sponsored or promoted.

work_type: Type of work associated with the job

currency: Currency in which the salary is provided.

compensation_type: Type of compensation for the job.

job_details/benefits.csv

job_id: The job ID

type: Type of benefit provided (401K, Medical Insurance, etc)

inferred: Whether the benefit was explicitly tagged or inferred through text by LinkedIn

company_details/companies.csv

company_id: The company ID as defined by LinkedIn

name: Company name

description: Company description

company_size: Company grouping based on number of employees (0 Smallest - 7 Largest)

country: Country of company headquarters.

state: State of company headquarters.

city: City of company headquarters.

zip_code: ZIP code of company's headquarters.

address: Address of company's headquarters

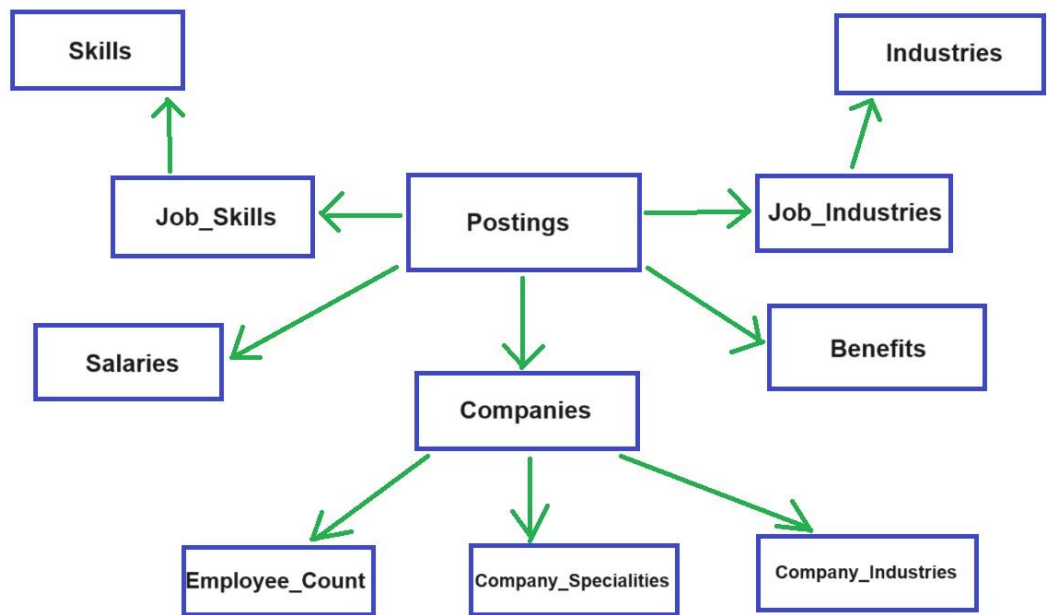
url: Link to company's LinkedIn page

company_details/employee_counts.csv

company_id: The company ID

employee_count: Number of employees at company
follower_count: Number of company followers on LinkedIn
time_recorded: Unix time of data collection

The relationships between these tables are as follows:



Data Quality Issues

Potential data quality issues that may arise include:

- **Incompleteness:** Some LinkedIn posts may lack essential information, such as job requirements or compensation details.
- **Inaccuracy:** Data provided by LinkedIn users may not always be accurate or up-to-date.
- **Inconsistency:** Data formats and terminology may vary across different LinkedIn posts.

Materials and Methods

First, I assessed the requirements of this information by asking frequently asked questions and understanding the needs of users, including companies and individual job seekers. The questions that emerged are as follows:

- Which jobs are remote?

- What benefits do the jobs offer?
- What is the payment structure for the jobs?
- What skills are required for the jobs?
- What type of work needs to be done for each job?
- What is the specialization of each company?
- Where are the companies located in terms of city, state, and country?

Then, based on these requirements, I divided them into two fact tables: Postings and Companies. The remaining data were divided into dimension tables.

The dimension tables for the Postings table include 6 tables: Benefits, Experience_Level, Pay_Period, Skills, Work_Type, and Company_Details. The dimension tables for the Companies table include 6 tables: Specialties, City, State, Country, Industries, and Company_Details.

The Company_Details table is a common table between the two fact tables. It is the primary key for the Companies table and the foreign key for the Postings table.

The fields for each of these tables are as follows:

Benefits Table:

- **id:** A unique primary key, auto-incremented for use as a foreign key in the Postings table.
- **type_name_set:** A set of benefit names for each job.

Work_Type Table:

- **id:** A unique primary key, auto-incremented for use as a foreign key in the Postings table.
- **type_name:** The type of work for each job.

Experience_Level Table:

- **id:** A unique primary key, auto-incremented for use as a foreign key in the Postings table.
- **level_name:** The name of the professional level required for each job.

Skills Table:

- **id:** A unique primary key, auto-incremented for use as a foreign key in the Postings table.
- **skill_abr_set:** An abbreviated name of each skill as a reference for acquiring the desired skill.
- **skill_name_set:** The full name of the skill required for each job.

Pay_Period Table:

- **id:** A unique primary key, auto-incremented for use as a foreign key in the Postings table.
- **period_name:** Describes the payment structure for each job.

Postings Table:

- **id:** The primary key for each job.
- **benefit_id:** A foreign key linking to the Benefits table.
- **skill_set_id:** A foreign key linking to the Skills table.
- **work_type_id:** A foreign key linking to the Work_Type table.
- **remote_allowed:** Indicates whether the job is remote or not.
- **experience_level_id:** A foreign key linking to the Experience_Level table.
- **pay_period_id:** A foreign key linking to the Pay_Period table.
- **expiry:** The expiration date of the job listing.
- **listed_time:** The date the job was listed.
- **company_id:** A foreign key linking to the Company_Details table.
- **views:** The number of views the job posting has received.
- **applies:** The number of applications submitted for the job.
- **max_salary:** The maximum salary for the job.
- **med_salary:** The median salary for the job.
- **min_salary:** The minimum salary for the job.

Industries Table:

- **id:** A unique primary key, auto-incremented for use as a foreign key in the Postings table.
- **industry_name:** The name of the industry the company operates in.

Specialities Table:

- **id**: A unique primary key, auto-incremented for use as a foreign key in the Postings table.
- **speciality_name_set**: A set of specializations each company has.

City Table:

- **id**: A unique primary key, auto-incremented for use as a foreign key in the Postings table.
- **city_name**: The name of the city where the company is located.

State Table:

- **id**: A unique primary key, auto-incremented for use as a foreign key in the Postings table.
- **state_name**: The name of the state where the company is located.

Country Table:

- **id**: A unique primary key, auto-incremented for use as a foreign key in the Postings table.
- **country_name**: The name of the country where the company is located.

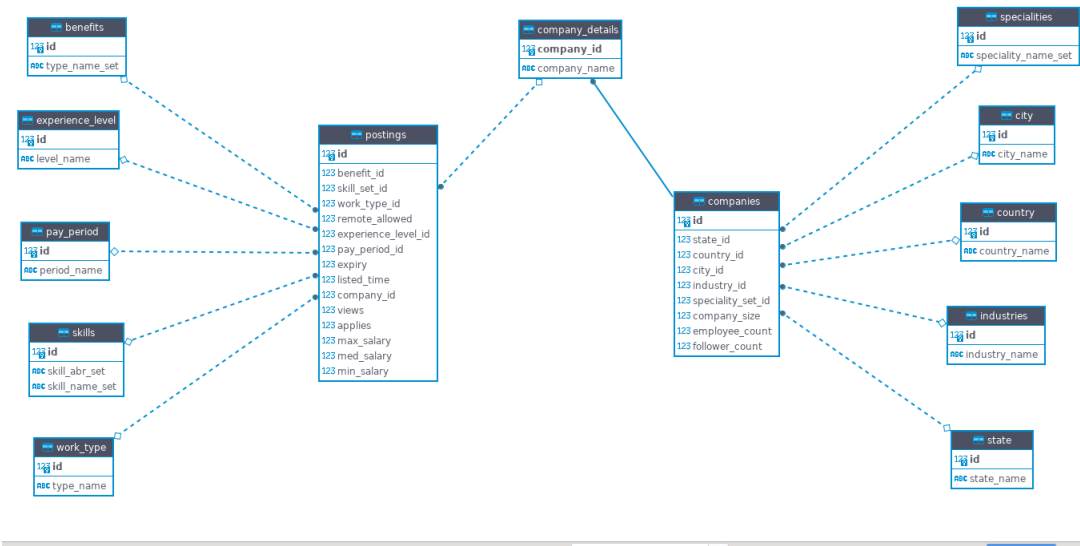
Companies Table:

- **id**: The primary key for each company, which serves as the primary key in the Company_Details table.
- **industry_id**: A foreign key linking to the Industries table.
- **speciality_set_id**: A foreign key linking to the Specialities table.
- **country_id**: A foreign key linking to the Country table.
- **state_id**: A foreign key linking to the State table.
- **city_id**: A foreign key linking to the City table.
- **company_size**: The calculated size of each company.
- **employee_count**: The number of employees in the company.
- **follower_count**: The number of followers the company has on LinkedIn.

Company_Details Table:

- **company_id**: The ID of the company, which serves as the primary key.
- **company_name**: The name of the company

The overall schema design of the data warehouse tables is as follows:



Workflow

To implement the ETL process for this project, I initially created the dimension tables using Pentaho. First, I read the CSV files corresponding to the dimension tables through Pentaho and loaded them into the data warehouse. Then, I implemented the Companies fact table, as it is necessary to load this table into the data warehouse before implementing the Postings fact table and the Company_Details table. After that, I implemented the Company_Details table, and finally, the Postings table was created. The overall workflow of this project, which is implemented as a job, is as follows:



LoadDimensions Implementation:

Here, all the dimension tables related to both Postings and Companies are implemented, except for one table, which requires the Companies fact table to be implemented first. The process of implementing the dimension tables is as follows:

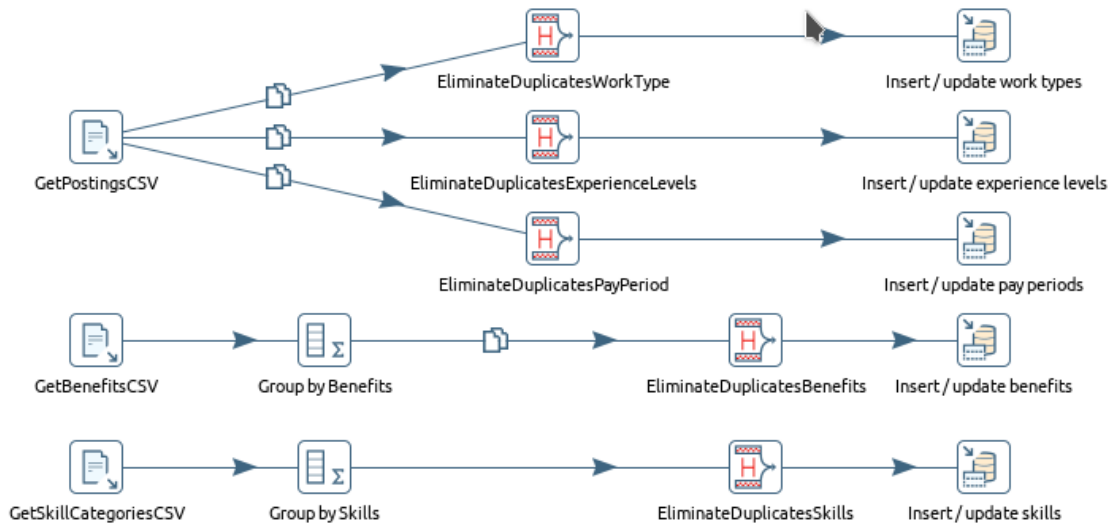


Image Description:

For the 'work_type', 'experience_level', and 'pay_period', all these attributes are present in the main 'postings' file and are repeated for every job. To avoid this repetition, I created a dedicated table for each attribute, assigned a unique row for each in their respective tables, and inserted them accordingly. Henceforth, only their IDs will be inserted into the fact table. For 'benefits' and 'skills', since each job must have a unique set of these attributes, I used the group by clause to create and insert this set for each job.

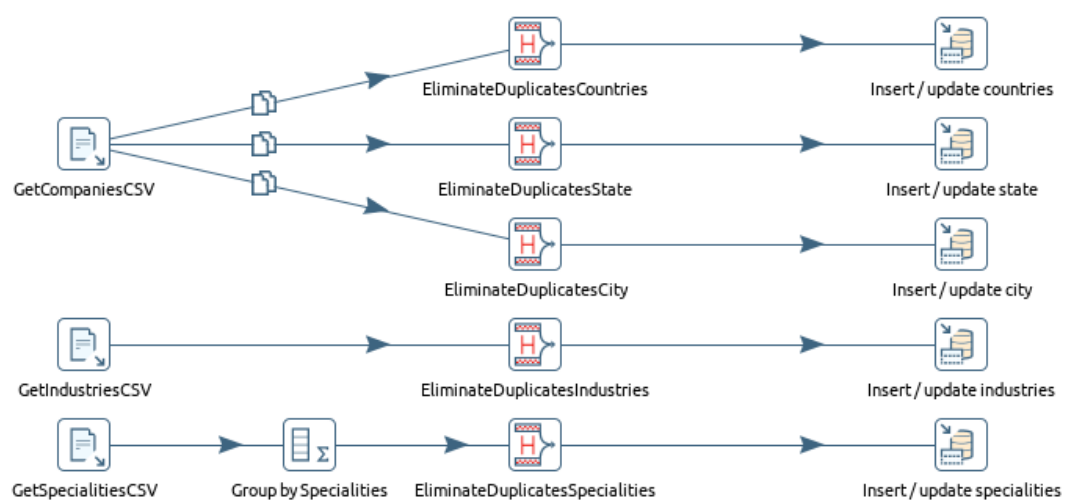


Image Description:

For the city, state, and country tables, all these values are present in the main companies file and are repeated in every row. To avoid this repetition, I assigned a unique table for each attribute and inserted each value into its respective table, assigning an ID to each. From now on, only their IDs will be inserted into the fact table.

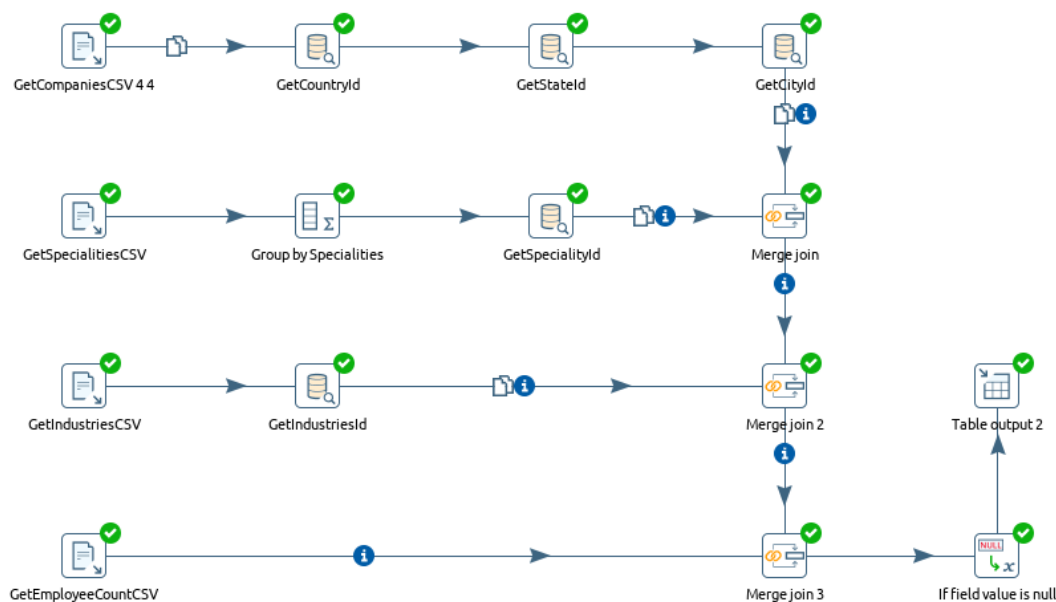
For the industries, each one is assigned to a unique row in its own table. Here, each industry is associated with a company.

For the specialities, since each company must have a unique set of specialities stored in the table, I used the group by clause to group these specialities and insert them into the data warehouse.

LoadFactCompanies Implementation:

In this process, I retrieve the necessary IDs from the data stored in the dimension tables within the data warehouse and insert them into the Companies fact table. Here's how I proceeded:

- 1. In the first step, I match the city, state, and country names from the CSV file and retrieve their respective IDs from their tables.
- 2. In the second step, I join the information obtained from the first step with the set of benefits, retrieving their IDs in a similar manner.
- 3. In the third step, I retrieve the IDs for the set of skills and join them with all the previously obtained information.
- 4. In the fourth step, I retrieve values for two fields from the Companies fact table.
- 5. Finally, I remove any null rows and insert the data into the Postings fact table. The process is as follows:



LoadRelationDimension Implementation:

This section is solely for populating the intermediate table between companies and postings. It is executed before loading the Postings fact table, as it is essential to reference it for the implementation of Postings. The process is as follows:



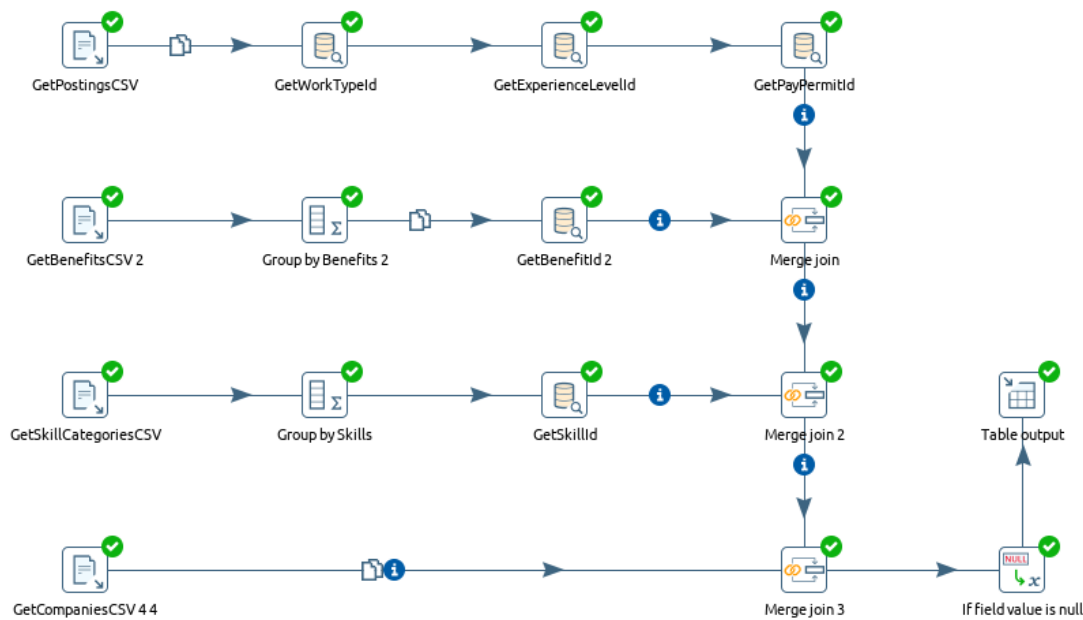
LoadFactPostings Implementation:

Here, as usual, I start by retrieving the IDs. In the first step, I match the field names in each of the 'work_type', 'experience_level', and 'pay_period' tables to retrieve their respective IDs.

In the second step, I use group by to construct a text representation of the benefits set for each job, matching it with what is stored in the 'benefits' table. I retrieve its ID and join it with the first step.

In the third step, I perform a similar operation with the skills, as in the previous step. In the fourth step, I join all this information with the Companies CSV file to obtain the 'company_id' for the Postings table.

Finally, I remove any null rows and insert the data into the Postings fact table. The process is as follows:



Software and Tools

The project will utilize the following software and tools:

- **Data Extraction Tool:** LinkedIn API library (LinkedIn API Library (Gathered and Acquired from Kaggle Website))
- **Data Transformation and Cleaning Tool:** Pentaho - Data Integration
- **Data Warehouse Management System:** PostgreSQL
- **Data Warehouse Reporting Tool:** Tableau

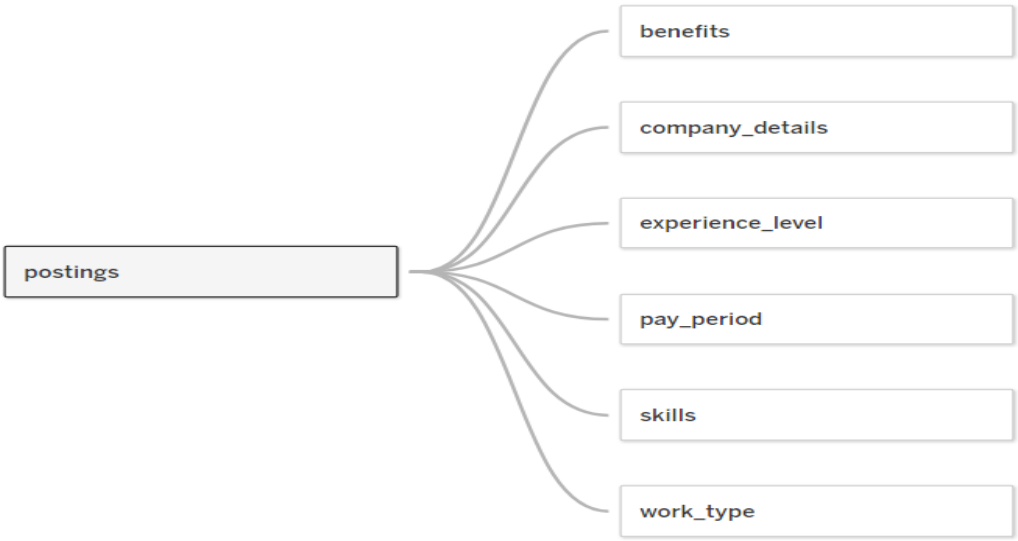
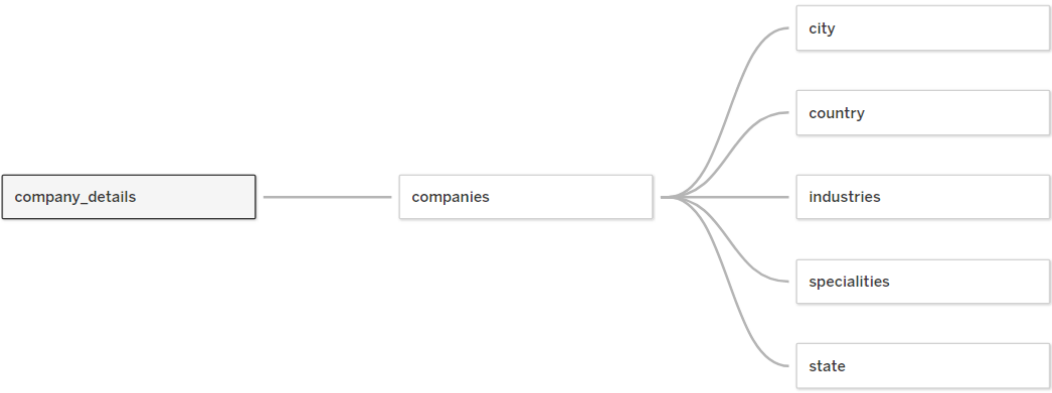
Results and Discussion

Data Statistics

- The extracted data includes 11 tables: 2 main tables and 9 dimension tables.
- The 9 tables that serve as dimension tables actually referred to the primary keys of the main tables.
- The main posts table has approximately 124,000 records, and the companies table has approximately 25,000 records.
- The data size for extraction is between 530 and 550 megabytes.
- The number and type of data warehouse tables changed to 13 different tables.

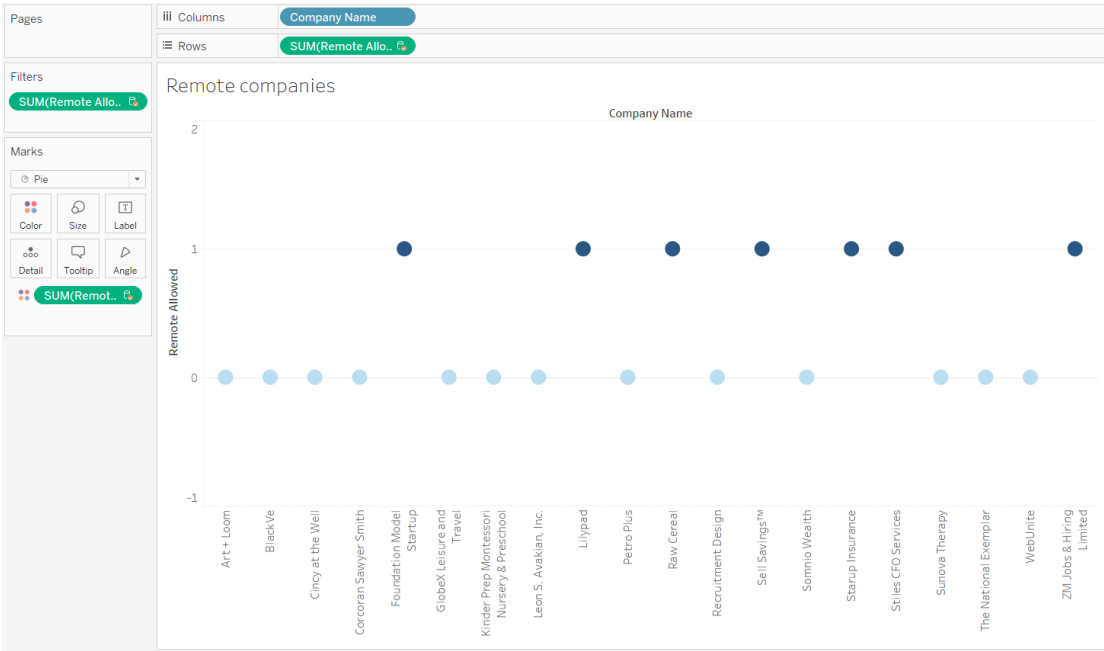
Analytical Queries and Results

These data can generate countless reports that can be extremely useful for regular users, various companies, as well as researchers and journalists, meeting their needs in the best possible way. I implemented several examples of these reports based on different criteria through Tableau software, which I will elaborate on further.



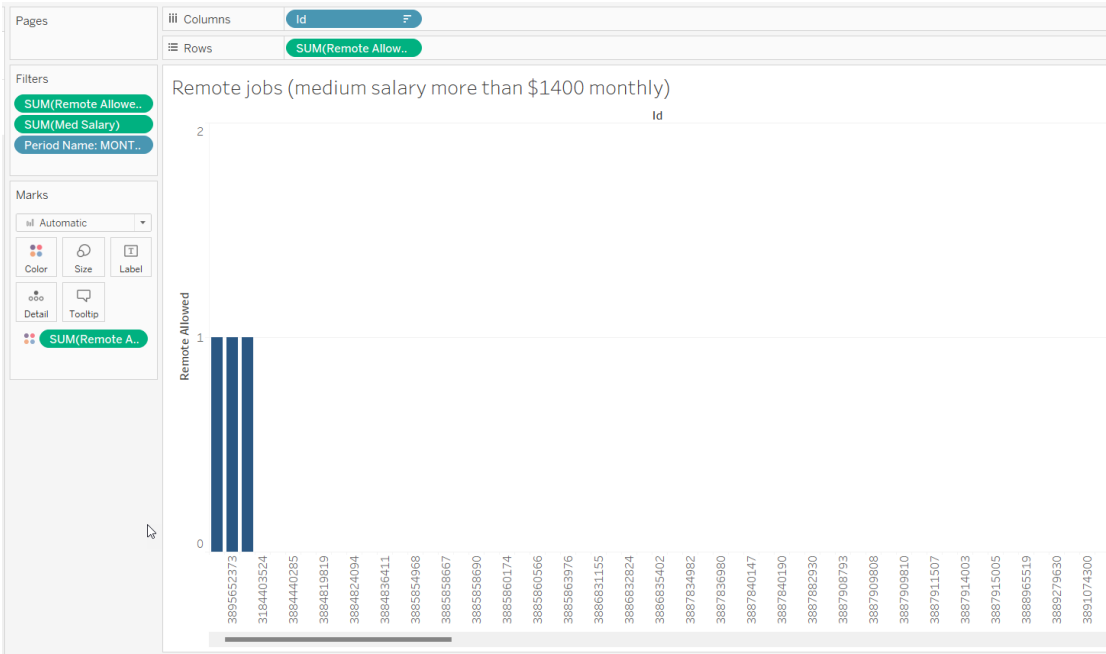
1- Which companies offer remote work services?

The first report is for users who are looking for jobs in their favorite companies but cannot work there in person if they are requested and accepted.



2- Which jobs are remote and have a monthly income of over \$1400?

The next report is for users who are specifically looking for remote work. These users either prefer remote work or cannot work in person. Additionally, the salary they are looking for is more than \$1400 per month.



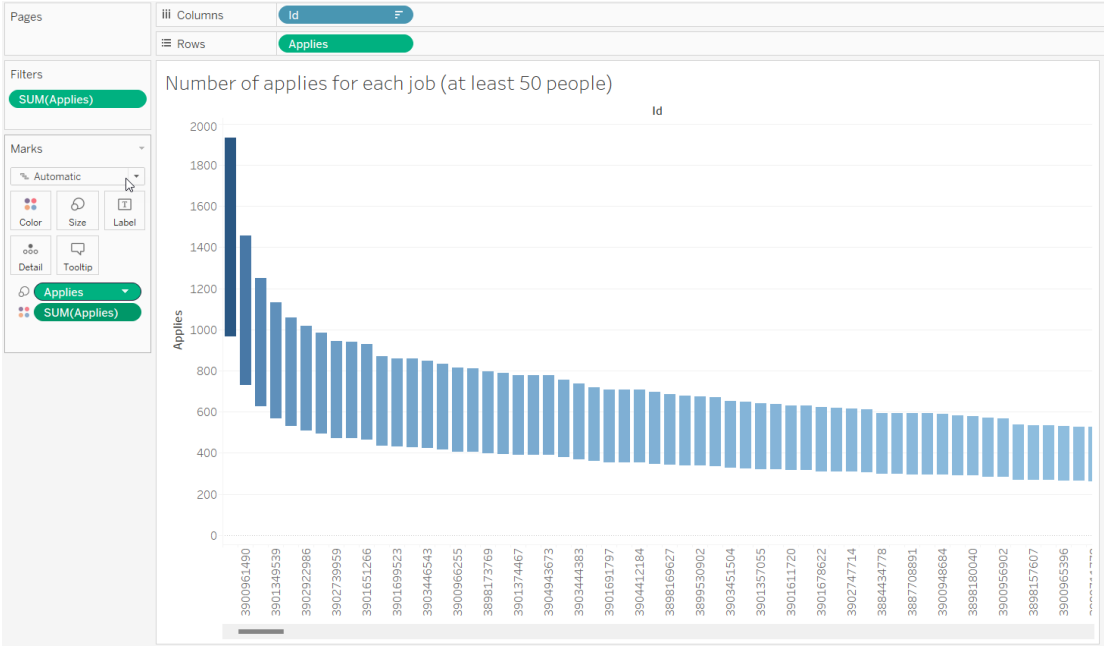
3- How do registered jobs contract with their employees?

This report is mainly for researchers and journalists who want to have reports beyond hiring, and purely for obtaining statistics to see which direction companies tend to cooperate with their employees.



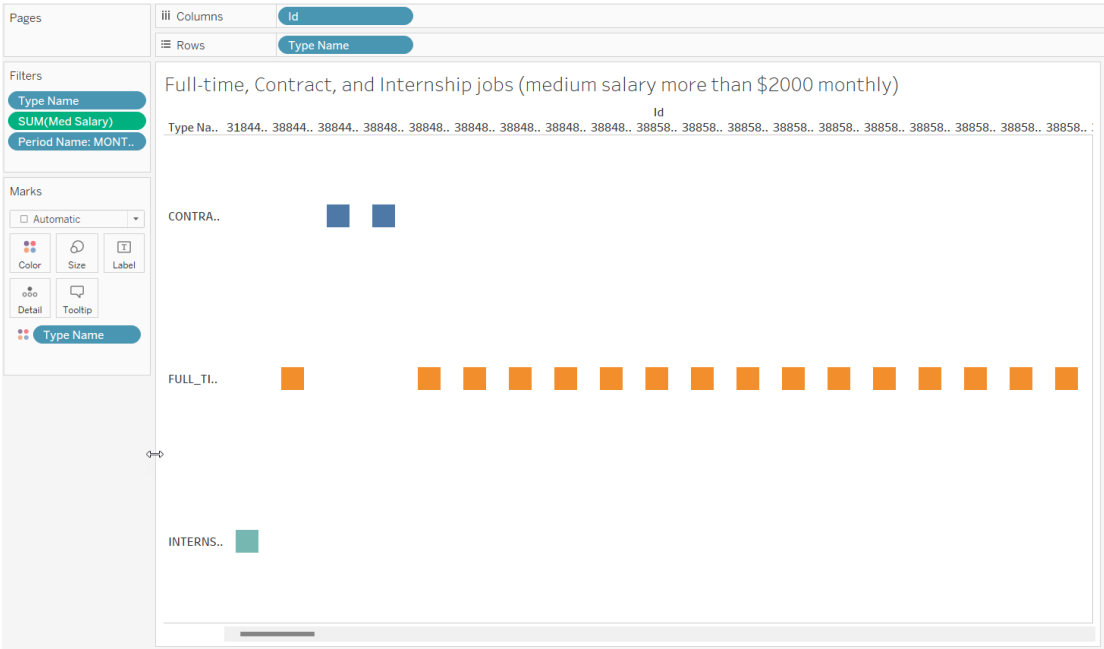
4- Which jobs have had more than 50 applicants?

In this report, I focused more on emotions, targeting users who are looking for jobs. Sometimes users might be looking for jobs with fewer applicants to avoid tough competition. Other times, users might be looking for jobs with many applicants to determine that the job might be valuable.



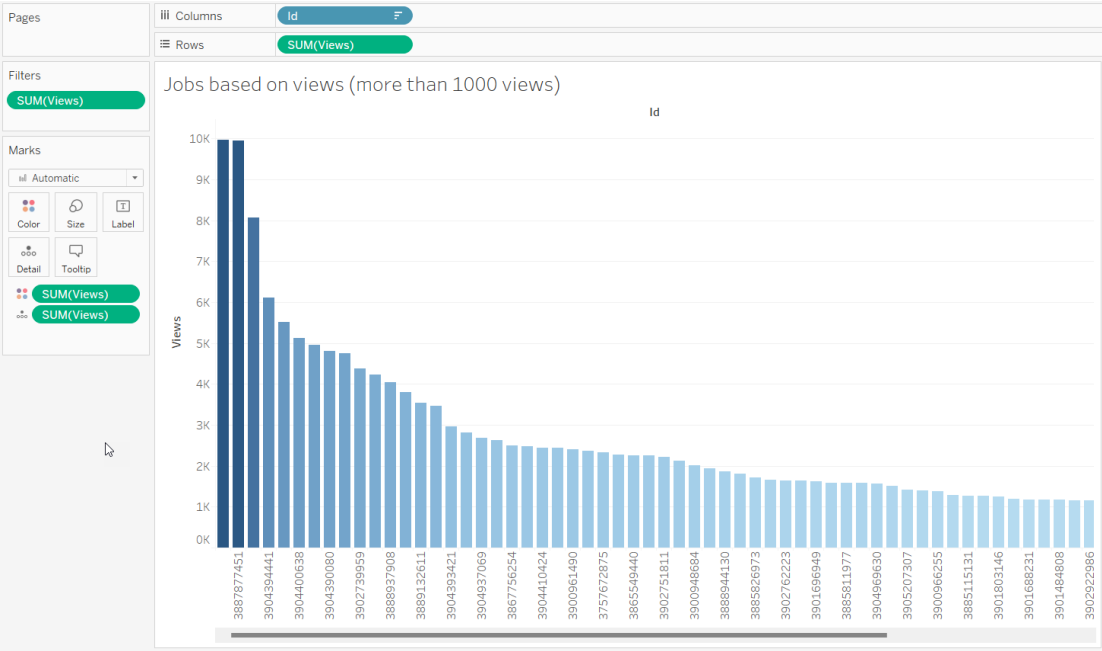
5- Which jobs are full-time, contract-based, or internships, and offer a salary of over \$2000 per month?

This report is for users who are looking for these three types of jobs and want a salary of over \$2000.



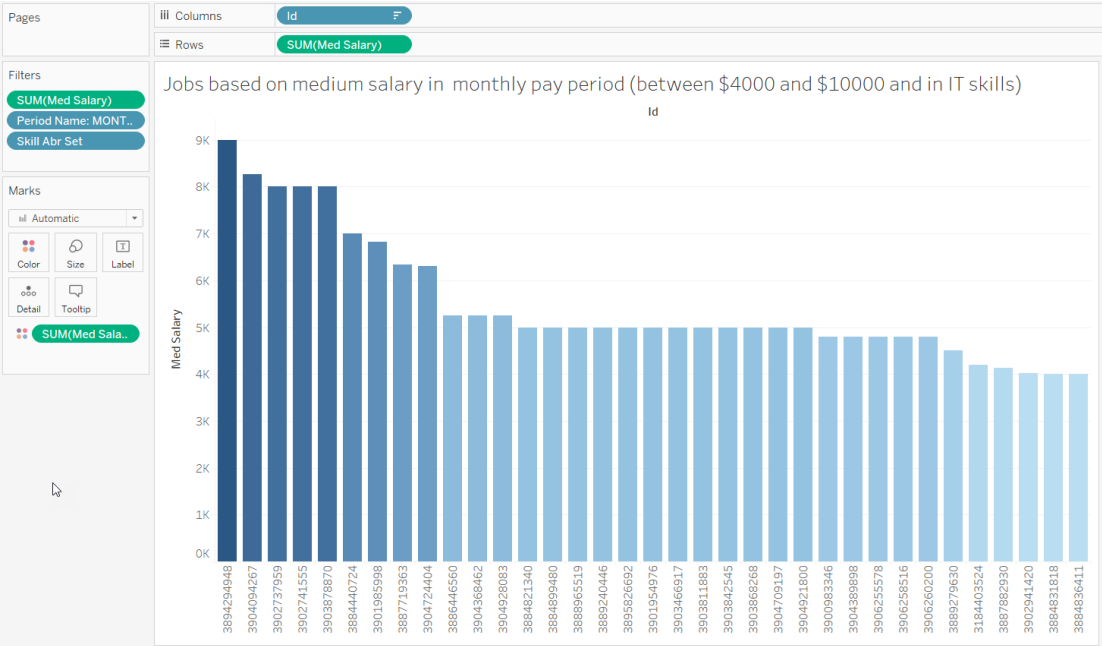
6- Which jobs have been viewed more than 1000 times?

This report focuses more on companies, researchers, and journalists. There might be situations where companies need to know which jobs attract more attention from users to guide their hiring process, and researchers to know which jobs people prefer more.



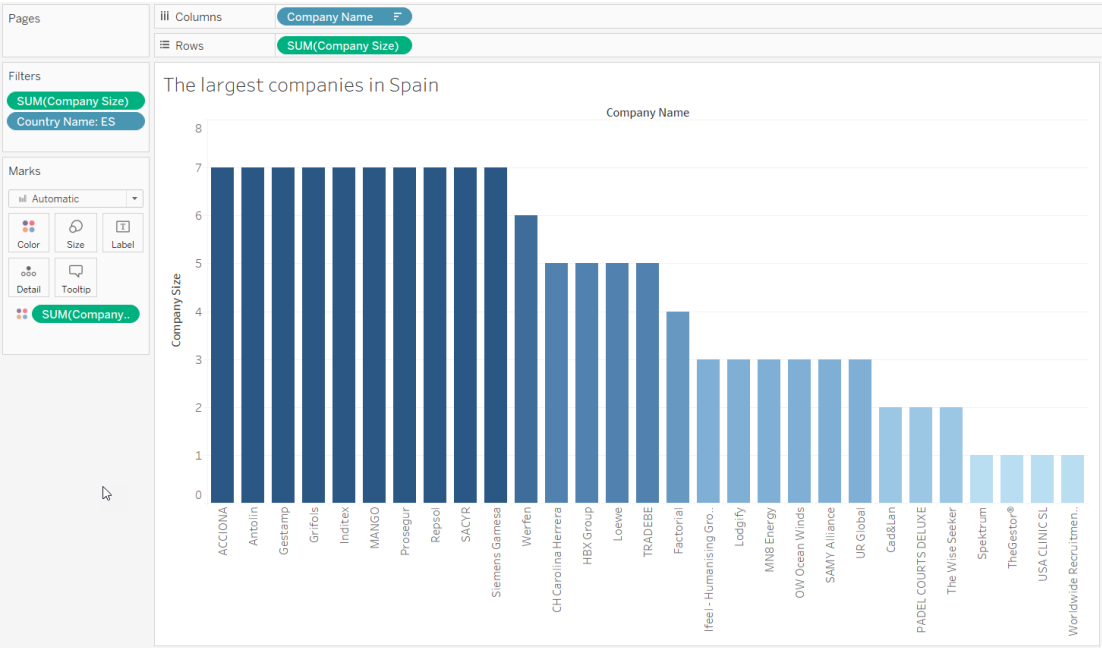
7- Which jobs have a monthly income between \$4000 and \$10000?

These jobs are usually for users who work professionally in the IT services field and are looking for high-income jobs between \$4000 and \$10000 per month.



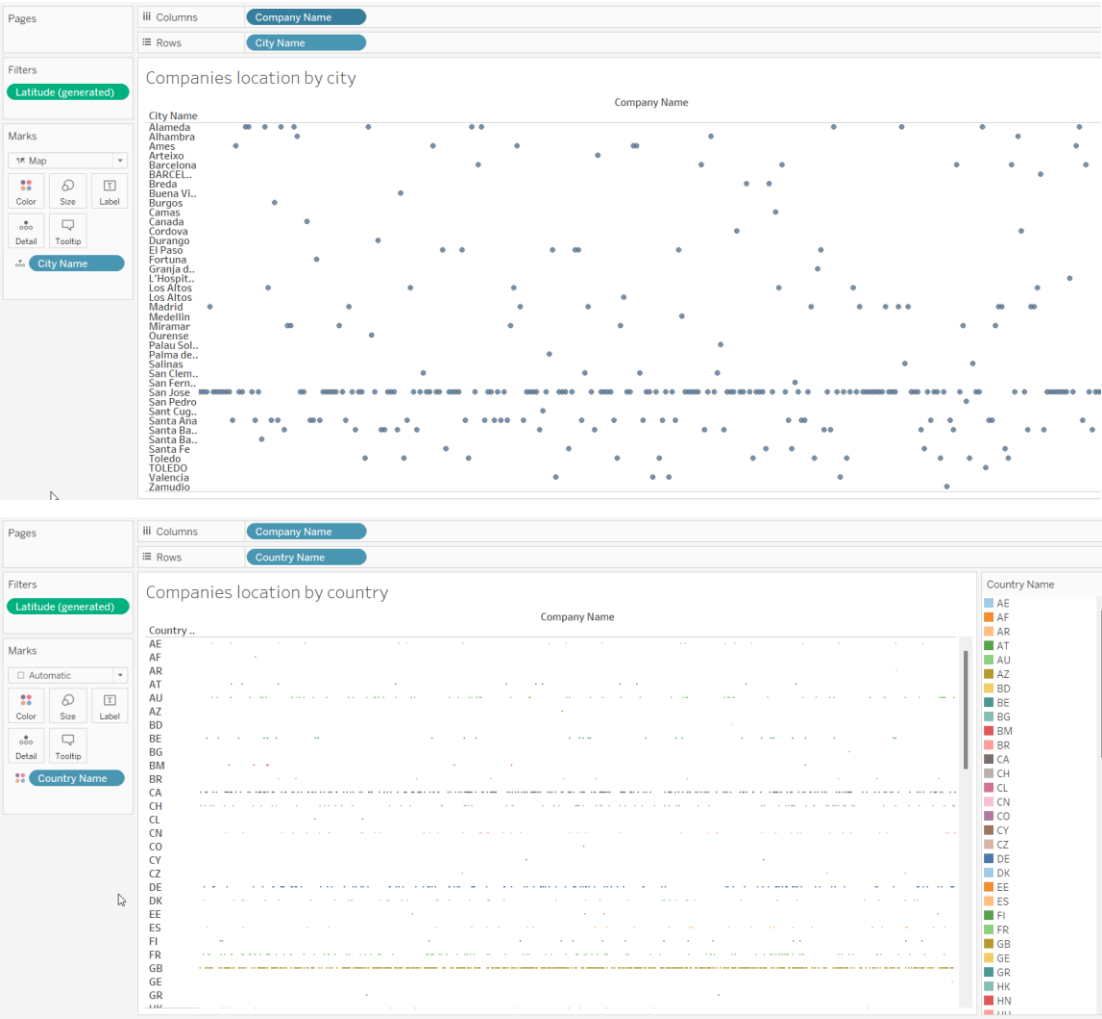
8- Which are the largest companies in Spain?

This report is mostly useful for researchers and journalists, who use it for some statistical reports to compare with other companies or for any other purpose.



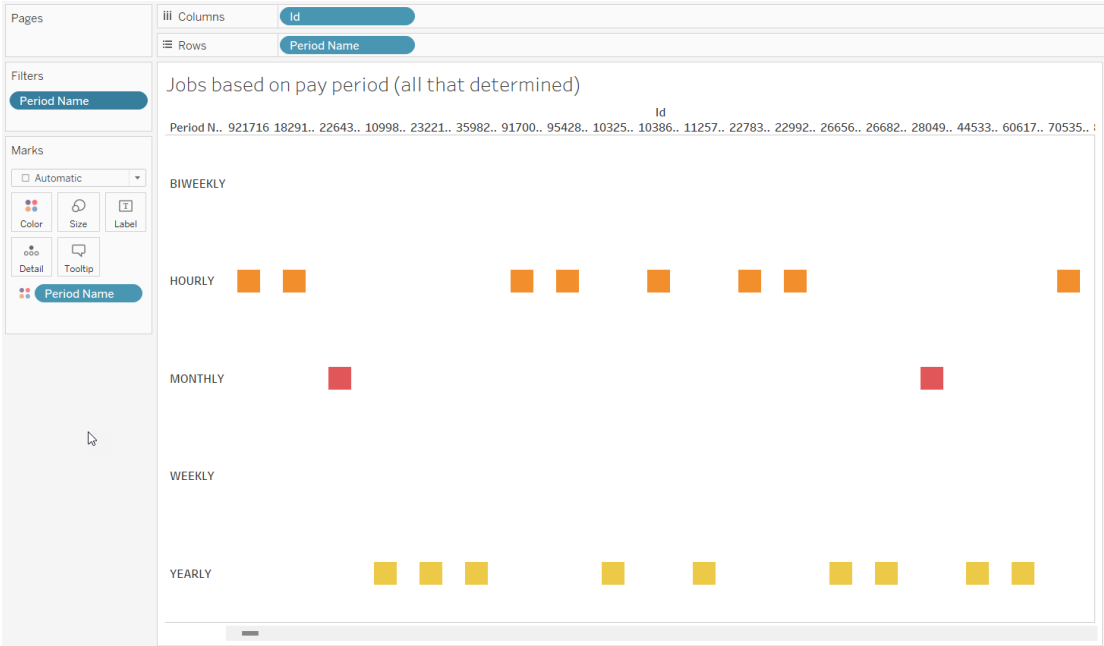
9- How do companies vary based on city or country?

These two reports are used by users and journalists. For users, to know in which city or country the company they want to apply for is located, and for journalists, to provide a report on how much need for hiring is felt in each city or country to examine the economic vitality of that region in the future.



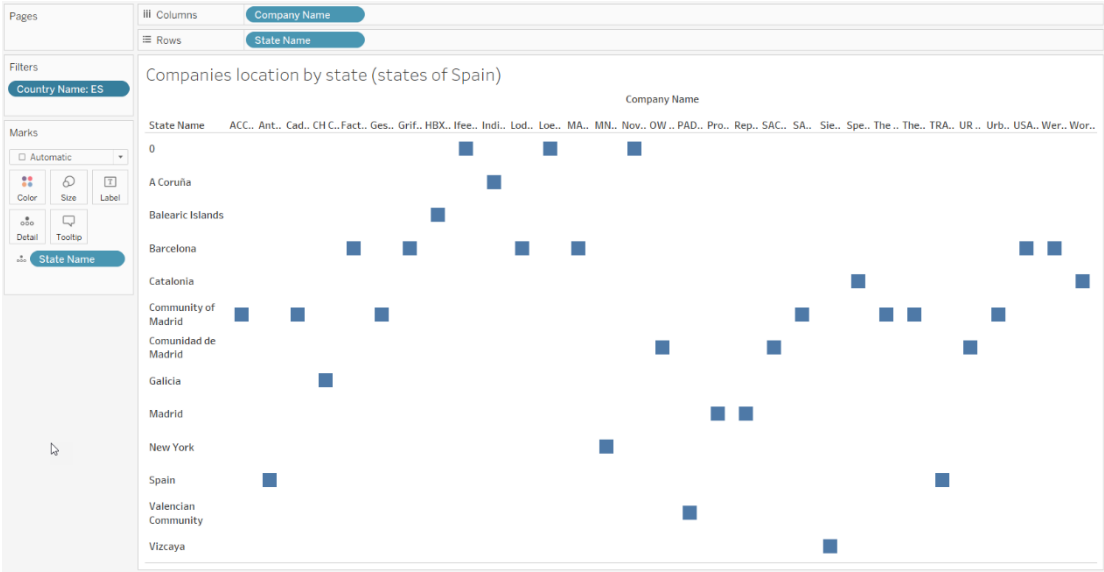
10- How are jobs based on the method of salary payment?

Sometimes companies need to model their salary payment methods after other companies and pay their employees more standardly. This report can provide a good perspective for them.



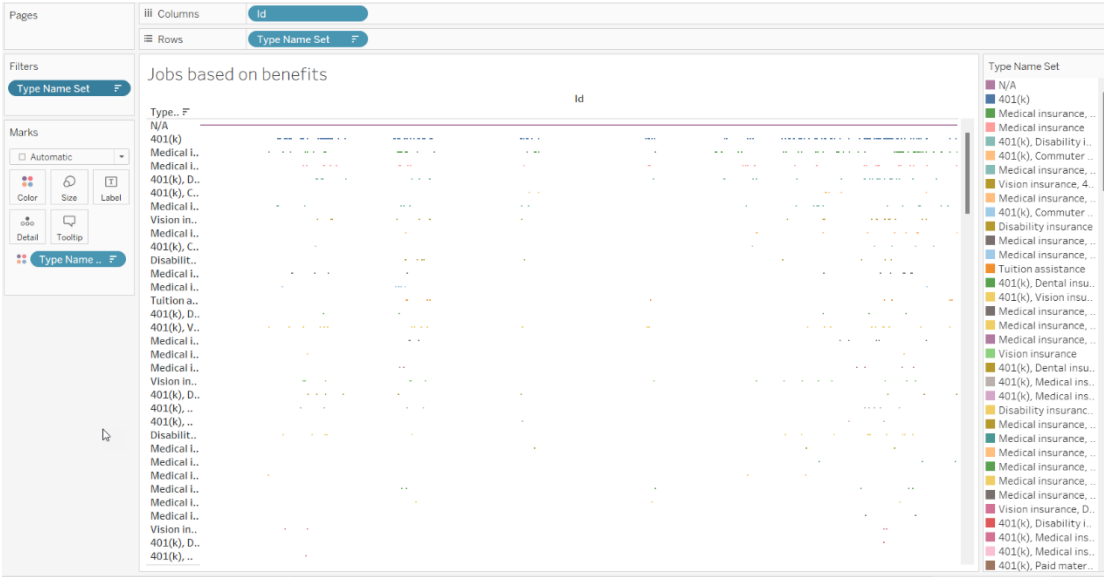
11- Which companies are in Spain?

This report is mainly used by researchers. They need to know the number and types of companies in Spain to have more accurate statistics in their reports.



12- What benefits does each job offer?

This report can be useful for all searchers. It is useful for users because they want to know what benefits and facilities the company they are applying to offers. For companies, to know whether the benefits they offer are acceptable in the hiring community or not. And for researchers and journalists to know which companies that offer these benefits have contracts with which other companies.



Discussion

The implemented ETL process and data warehouse effectively address the project's analytical goals, providing a centralized repository of high-quality data for informed decision-making. The project highlights the importance of data quality management and data modeling in ensuring the success of data warehousing initiatives.

Conclusion

The project successfully demonstrates the implementation of an ETL process and the establishment of a data warehouse for LinkedIn post analysis. The project provides a valuable framework for organizations seeking to leverage LinkedIn data for strategic advantage.

References

<https://www.youtube.com/watch?v=8dGabtBxa4Y>

<https://www.youtube.com/watch?v=-heDV0ZwR3w&list=PLT8E937iDWrzxeYT5iZbiko0KWb2CL8lq&index=5>

<https://www.youtube.com/@ynteractivetraining/playlists>

https://www.youtube.com/watch?v=9uBtK6j_QgA&list=PLoyECfvEFOjYXcuVCBVd2hoDPAr83suuK